

CO-1-AT-1

1. A factory has three boxes containing colored balls. Box A contains 6 red balls; Box B contains 5 red 7 blue and 8 green balls; and Box C contains 8 red, 2 blue and 5 green balls. One box is selected at random, but box B is twice as likely to be chosen as each of the other boxes. A ball drawn from the chosen box is found to be blue. What is the probability that the selected box was box C?

Given

$$A = 1 \text{ part} \quad B = 2 \text{ part} \quad C = 1 \text{ part}$$

$$\text{Total} = 4 \text{ parts}$$

$$P(A) = \frac{1}{4}, \quad P(B) = \frac{1}{2}, \quad P(C) = \frac{1}{4}$$

$$P(\text{Blue}/A) = \frac{4}{12} = \frac{1}{3}$$

$$P(\text{Blue}/B) = \frac{7}{15}$$

$$P(\text{Blue}/C) = \frac{2}{15}$$

Using Bayes' Theorem:

$$P(C|\text{Blue}) = \frac{P(C) P(\text{Blue}/C)}{P(A) P(\text{Blue}/A) + P(B) P(\text{Blue}/B) + P(C) P(\text{Blue}/C)}$$

$$= \frac{\frac{1}{4} \cdot \frac{2}{15}}{\frac{1}{4} \cdot \frac{1}{3} + \frac{1}{2} \cdot \frac{7}{15} + \frac{1}{4} \cdot \frac{2}{15}} = \frac{\frac{1}{30}}{\frac{1}{12} + \frac{7}{15}} = \frac{8}{31}$$

$$\therefore P(C|\text{Blue}) = \frac{2}{31} \approx 6.45\%$$

3

Design a machine learning system for disaster management to predict events such as floods and earthquakes using historical and real time data. Explain the choice of model, data preparation and feature selection techniques

(a) Describe the steps involved in training, validating and evaluating the model

(b) Explain how the system can assist in early warning and emergency response

Given

$$P(A) = \frac{1}{4}, P(B) = \frac{1}{2}, P(C) = \frac{1}{4}$$

$$P(\text{Blue}/A) = \frac{3}{10}, P(\text{Blue}/B) = \frac{6}{15} = \frac{2}{5}$$

$$P(\text{Blue}/C) = \frac{2}{15}$$

Using Bayes Theorem

$$\begin{aligned} P(C/\text{Blue}) &= \frac{\frac{1}{4} \cdot \frac{2}{15}}{\frac{1}{4} \cdot \frac{3}{10} + \frac{1}{2} \cdot \frac{2}{5} + \frac{1}{4} \cdot \frac{2}{15}} \\ &= \frac{1/30}{3/40 + 1/5 + 1/30} \\ &= \frac{4}{9+24+4} \\ &= \frac{4}{37} \end{aligned}$$

$$\therefore P(C/\text{Blue}) = \frac{4}{37} \approx 10.81\%$$

3. In a real world medical image classification
 A deep learning model with 12000 is used to
 classify breast cancer as malignant or benign.
 The dataset has 1800 samples per class with
 20% used for testing. The evaluation gives
 $TP = 210$, $TN = 190$, $FP = 30$. Find the FN
 and calculate Accuracy, Precision, Sensitivity
 (Recall), and specificity. Comment on the model's
 effectiveness in medical diagnosis

Total dataset: $1800 + 1800 = 3600$

20% testing data: $3600 \times 20\% = 720$

Given:

$TP = 210$ $TN = 190$ $FP = 30$ $FN = ?$

Therefore $FN = 720 - (210 + 190 + 30)$

$FN = 290$

$$\text{Accuracy} = \frac{TP + TN}{TP + TN + FP + FN} = \frac{210 + 190}{720}$$

$$= 55.56\%$$

$$\text{Precision} = \frac{TP}{TP + FP} = \frac{210}{210 + 30} = 42\%$$

$$\text{Specificity} = \frac{TN}{TN + FP} = \frac{190}{190 + 30} = 86.36\%$$

Thus it is not sufficiently reliable for medical
 diagnosis without improvement.

4. Examine the performance of ML model using its training and validation loss values over multiple epochs:

- (a) Epoch 1: Training loss = 2.8, Validation loss = 2.4
- (b) Epoch 2: Training loss = 2.2, Validation loss = 2.2
- (c) Epoch 3: Training loss = 2.2, Validation loss = 2.0
- (d) Epoch 4: Training loss = 2.9, Validation loss = 2.1
- (e) Epoch 5: Training loss = 1.8, Validation loss = 2.3
- (f) Epoch 6: Training loss = 1.6, Validation loss = 2.6
- (g) Epoch 7: Training loss = 1.5, Validation loss = 2.9
- (h) Epoch 8: Training loss = 1.3, Validation loss = 3.2

- (i) Identify the epoch at which model begins to overfit
- ii) Suggest two effective techniques to reduce overfitting in this model
- iii) Discuss how overfitting affects the model's generalisation ability on unseen data.

i) Epoch where overfitting begins
Validation loss decreases until epoch 3 (2.0)
From epoch 4, training loss continues to decrease but validation loss starts increasing
Overfitting begins at Epoch 4

ii) Two techniques to reduce overfitting
a) Dropout - randomly disable some neurons during training; b) Early stopping - stops training when validation loss starts increasing

iii) Effects on generalisation: It performs well on training data but poorly on unseen/test data, reducing generalisation ability

5. Examine the performance of ML model using training and validation accuracy over multiple concepts.

Epoch	Training accuracy	validation accuracy
(a) 1	60%	63%
(b) 2	65%	68%
(c) 3	70%	72%
(d) 4	75%	73%
(e) 5	80%	72%
(f) 6	85%	70%
(g) 7	88%	68%
(h) 8	92%	

- Identify the Epoch at which the model starts overfitting.
- Explain why validation accuracy decreases despite training accuracy increasing.
- Suggest two methods to improve validation performance.

i) Epoch where overfitting starts: validation accuracy increases until Epoch 5 (73%) from epoch 6, training accuracy continues increasing while validation accuracy decreases overfitting starts at Epoch 6

ii) Training accuracy increases but performance on unseen validation data decreases

iii) Dropout/Regularization are two methods to early stopping validation performance